

WAZUH:Open Source Platform

CICSA PROJECT

submitted by

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To

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ACKNOWLEDGEMENT

First and foremost, I would like to express our thanks to Almighty for the diving grace bestowed on us to complete this project successfully on time and who have guided and supported me in bringing this project to completion.

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Once again We convey our gratitude to all those who had direct or indirect influence on our project.

ABSTRACT

Wazuh is an open-source security platform that provides comprehensive threat detection, incident response, and continuous monitoring for systems and networks. Acting as both a SIEM and XDR solution, it centralizes the collection, analysis, and visualization of security data from multiple sources such as servers, desktops, cloud environments, and containers. Operating on a client-server model, it includes agents installed on endpoints that collect logs and security data, a manager that analyzes the information using predefined rules to identify threats or policy violations, and an indexer that stores processed data for quick retrieval, all visualized through a web-based dashboard. Its key features include log analysis, file integrity monitoring, intrusion and vulnerability detection, security configuration assessment, and automated responses to security incidents. Wazuh collects data from agents, analyzes it with detection rules, generates alerts, and stores them for visualization. This enables administrators to detect threats like brute-force attacks, privilege escalations, and rootkits in real time. It can also trigger automated responses, such as blocking malicious IPs, making it a scalable, lightweight, and cost-effective end-to-end security solution.

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INTRODUCTION

Wazuh provides a pre-built virtual machine image in Open Virtual Appliance (OVA) format. This can be directly imported to VirtualBox or other OVA compatible virtualization systems. Take into account that this VM only runs on 64-bit systems with x86_64/AMD64 architecture. It does not provide high availability and scalability out of the box. However, these can be implemented by using distributed deployment.

Wazuh is an open-source security monitoring and threat detection platform designed to help organizations protect their systems from cyberattacks and policy violations. It collects, analyzes, and correlates security data from multiple sources, providing real-time alerts, log analysis, vulnerability detection, and compliance monitoring through a centralized dashboard.¹

The primary objective of Wazuh is to enhance system security by continuously monitoring activities across servers, endpoints, and networks. It identifies suspicious behavior, unauthorized access, configuration issues, and other security threats, enabling quick incident response and better visibility into the security posture of an organization.

Agents in Wazuh play a crucial role by collecting security logs and system data from endpoints such as Kali Linux, Windows, or macOS. When a Kali Linux machine is added as an agent, it sends system activity, file integrity, and security event logs to the Wazuh manager for analysis. Overall, Wazuh is important because it provides a powerful, automated, and centralized solution for detecting threats, managing compliance, and improving overall cybersecurity.

SYSTEM REQUIREMENTS

HARDWARE

System configuration:

Processor : Core i5-13450HX processor, an NVIDIA GeForce RTX 3050

RAM : 8GB

SOFTWARE

❖ WINDOWS 11

Windows 11 serves as the host operating system used to download the Wazuh OVA file and install Oracle VirtualBox. It provides the platform on which the virtual machines (Wazuh server and Kali Linux agent) are created and managed.

❖ VIRTUAL BOX

VirtualBox is a virtualization software used to import and run the Wazuh server OVA file. It allows the creation of isolated virtual environments where both the Wazuh server and the Kali Linux agent can operate simultaneously without affecting the host system.

❖ KALI LINUX

System configuration:

Operating System : Linux 2.6/3.x/4.x/5.x (64bit)

Processor : 4

Base Memory : 4048MB

Network : Intel PRO/1000 MT Desktop (Bridged Adapter)

❖ WAZUH

System configuration:

Operating System : Linux 2.6/3.x/4.x/5.x (64bit)

Processor : 4

Base Memory: 8192MB

Graphic controller : select the VBoxVGA

Network : Intel PRO/1000 MT Server (Bridged Adapter)

Version : v4.13.1

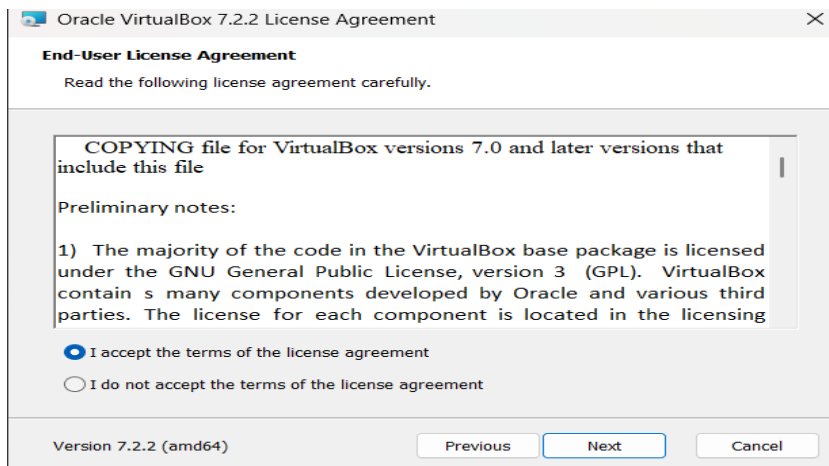
IMPLEMENTATION

1. ORACLE VIRTUALBOX INSTALLATION

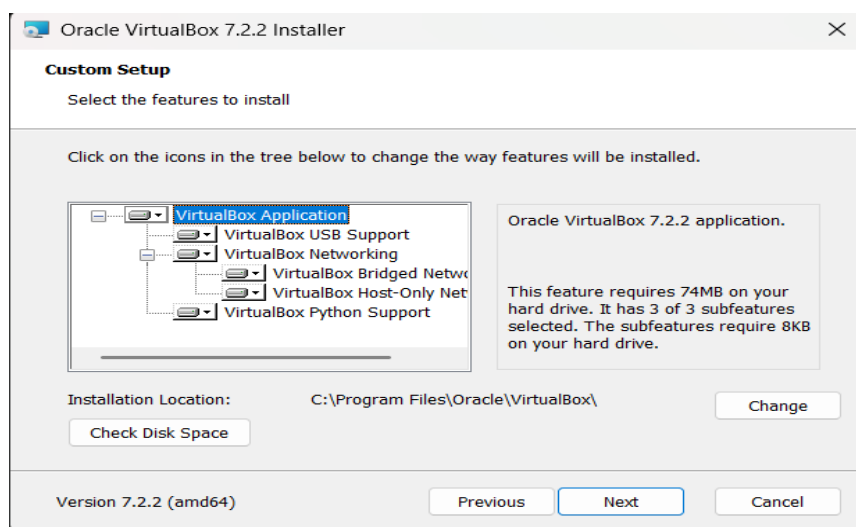
Step 1: Download the virtual box on windows by the following link;

<https://download.virtualbox.org/virtualbox/7.2.2/VirtualBox-7.2.2-170484-Win.exe>

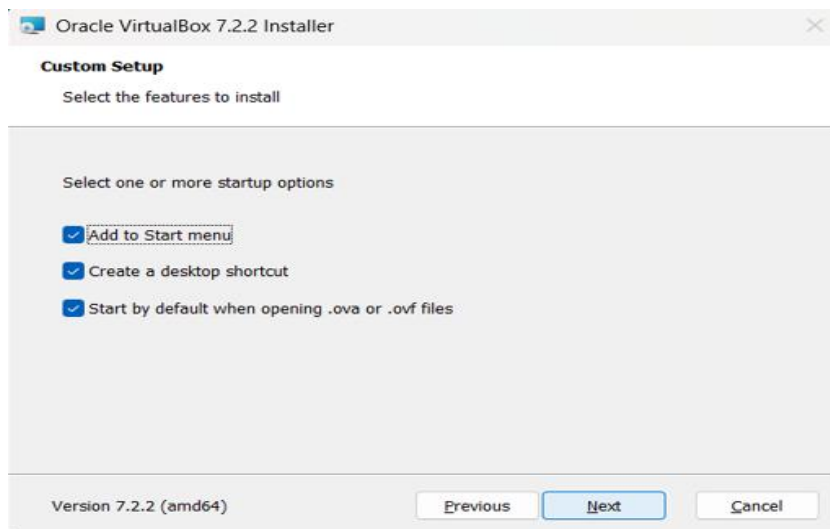
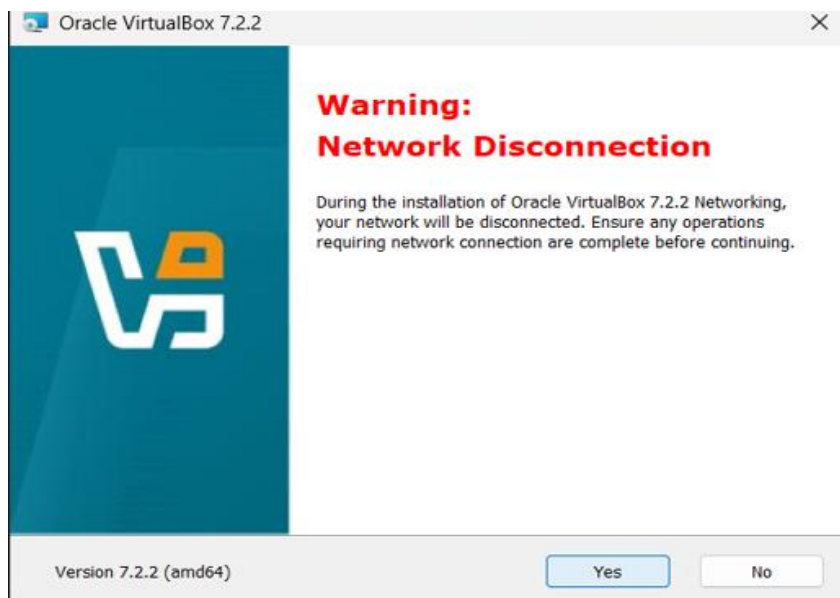
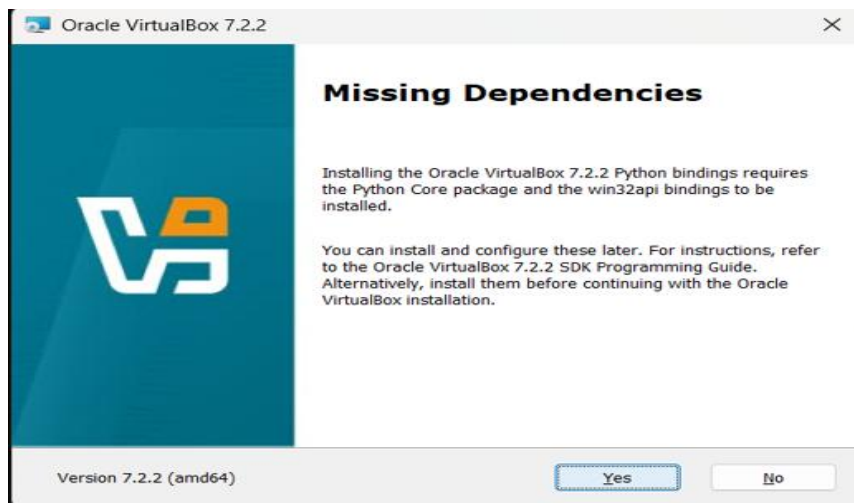
Step 2 : Accept the license agreement



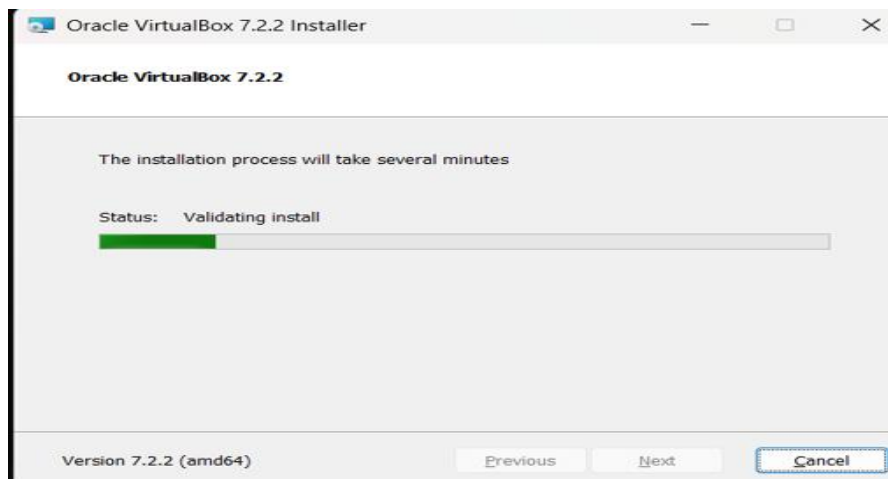
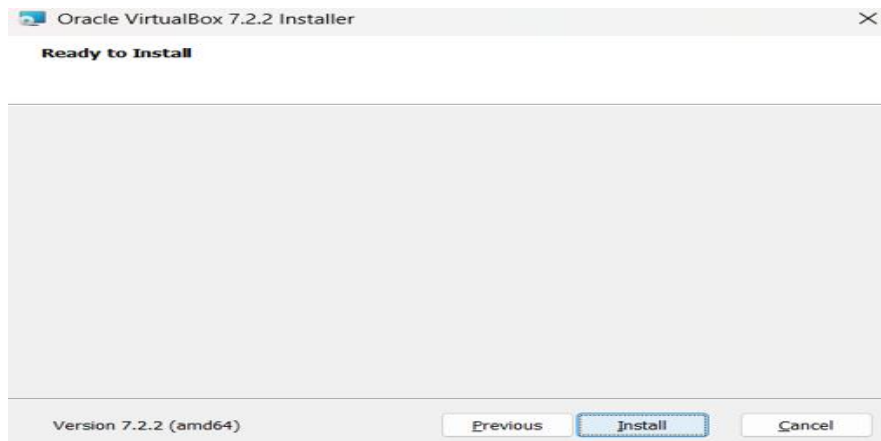
Step 3 : Select the feature to install.



Step 4 : click yes and continue



Step 5 : Click Install and wait for installation to complete



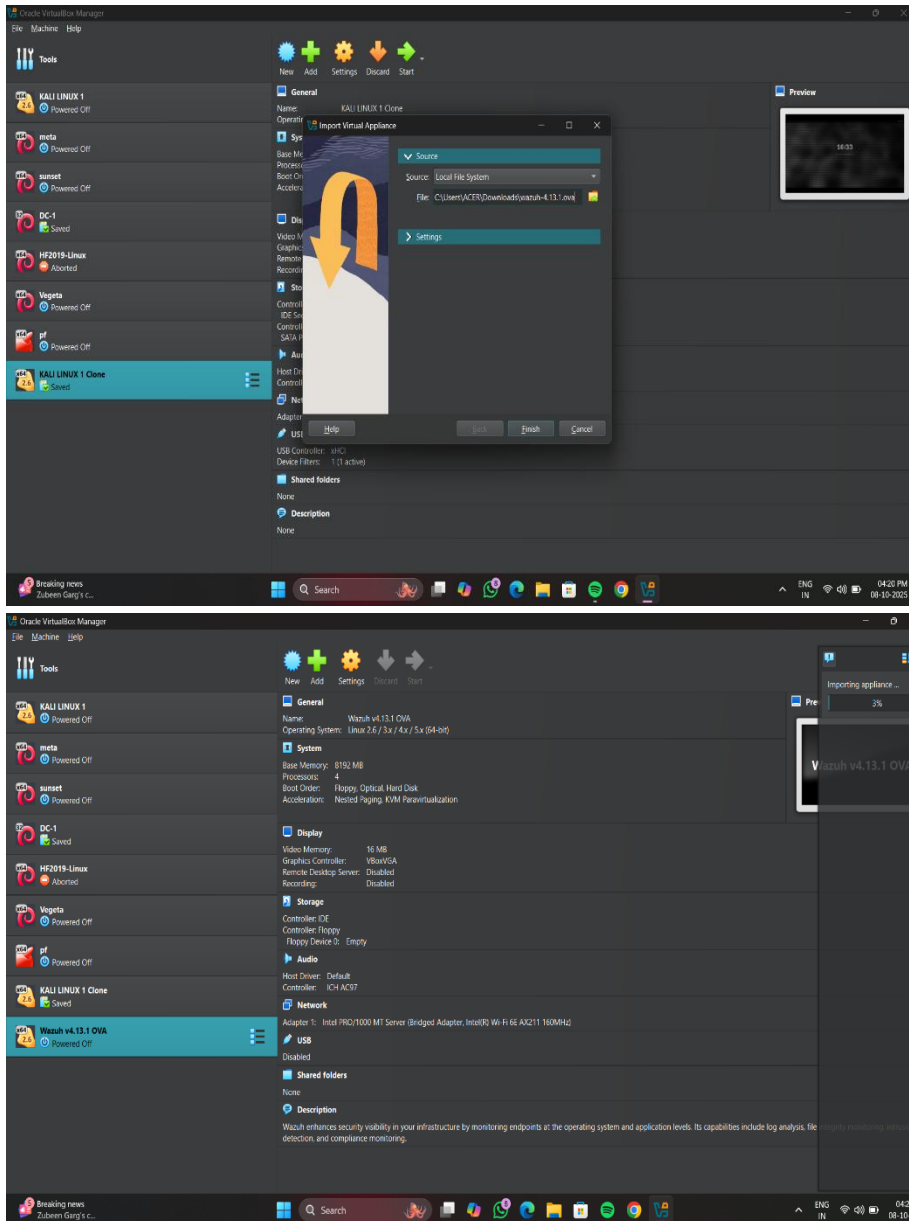
Step 6 : Installation completed , click finish



2. WAZUH INSTALLATION

Step 1 : Download the wazuh.ova file from the following link;
<https://packages.wazuh.com/4.x/vm/wazuh-4.13.1.ova>

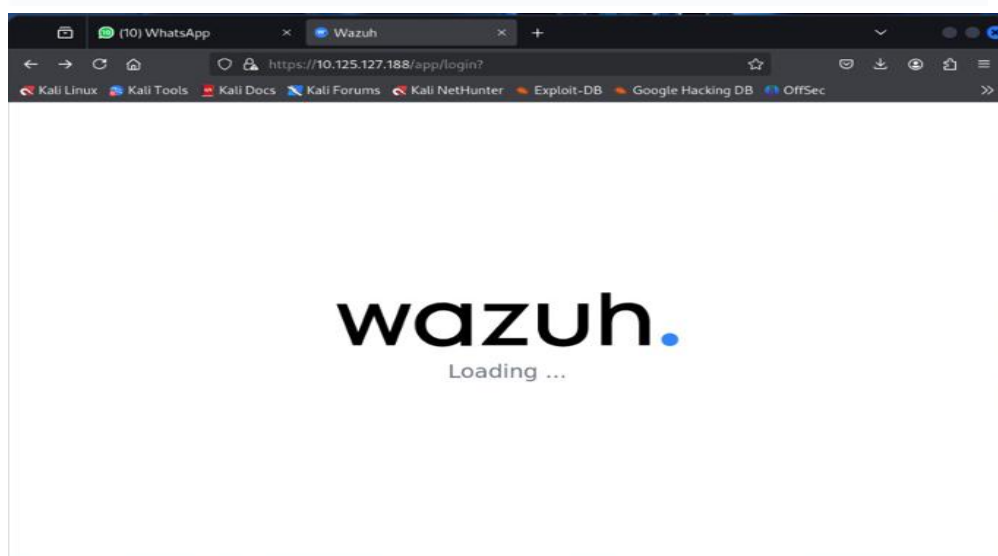
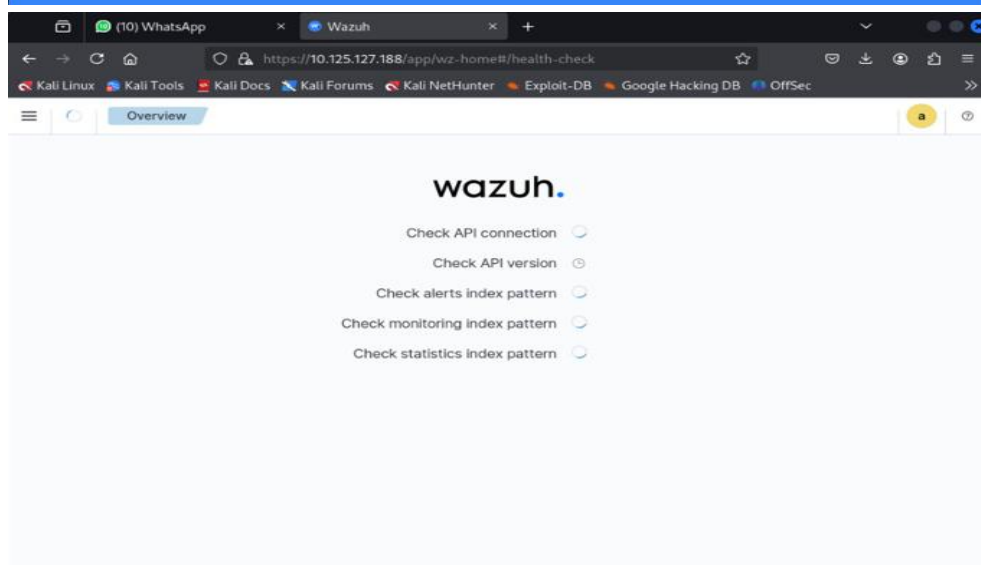
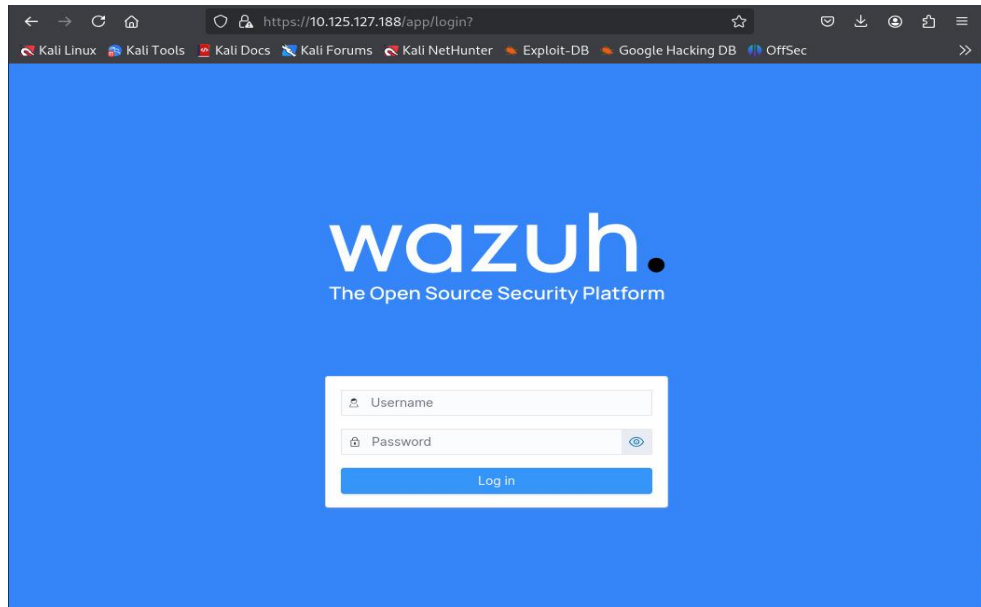
Step 2 : Open virtual box , File→import Appliances



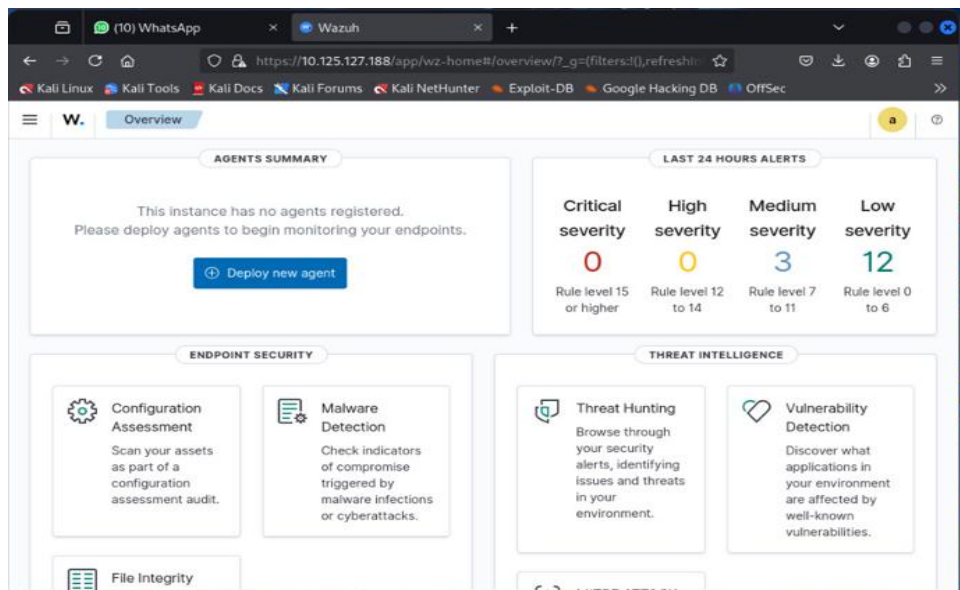
After importing the appliances finishes the wazuh-server is ready to perform

[illegible][illegible]

Step 5 : Using the server address login to the wazuh using
Username : admin and password : admin

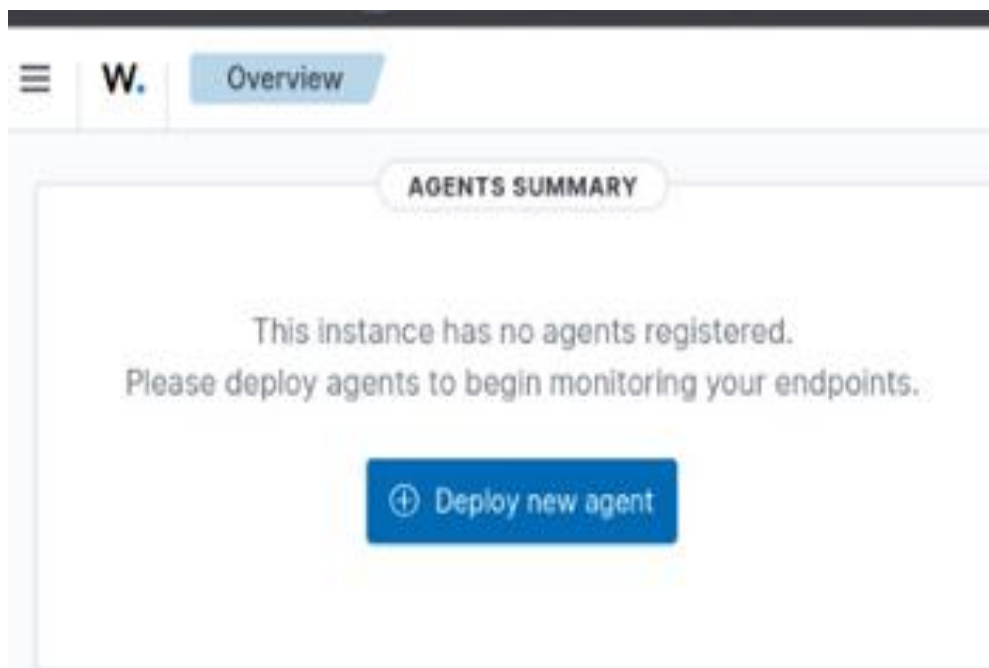


Successfully logged into the wazuh



3. WAZUH-AGENT DEPLOYMENT

Step 1 : click on Deploy new agent



Step 2 : Select the package DEB amd64 linux version and enter the server address

Deploy new agent

Select the package to download and install on your system:

LINUX

☐ RPM amd64 ☐ RPM aarch64

☒ DEB amd64 ☐ DEB aarch64

WINDOWS

☐ MSI 32/64 bits

macOS

☐ Intel ☐ Apple silicon

For additional systems and architectures, please check our [documentation](#).

Server address:

This is the address the agent uses to communicate with the server. Enter an IP address or a fully qualified domain name (FQDN).

Assign a server address

Step 3 : Enter the agent name : KaliLinux and assign Default

Optional settings:

By default, the deployment uses the hostname as the agent name. Optionally, you can use a different agent name in the field below.

Assign an agent name:

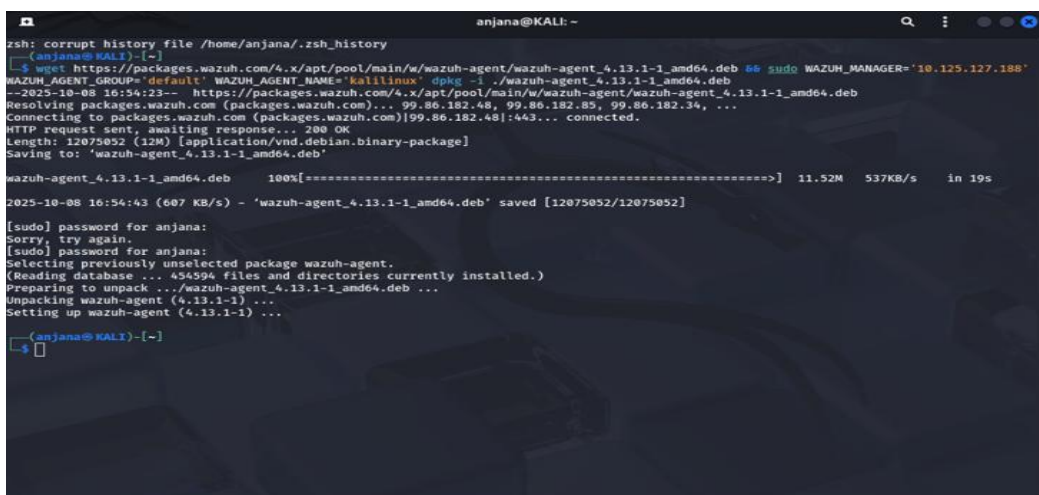
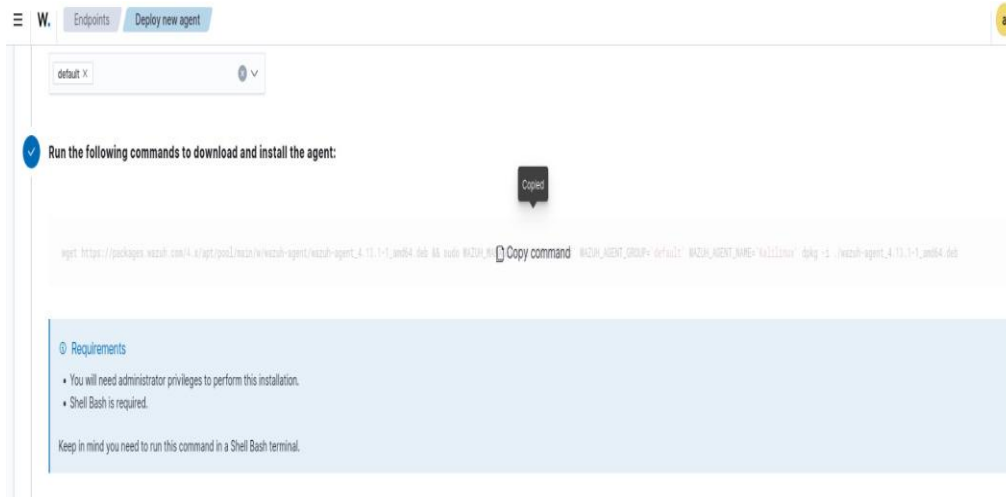
Kalilinux

The agent name must be unique. It can't be changed once the agent has been enrolled.

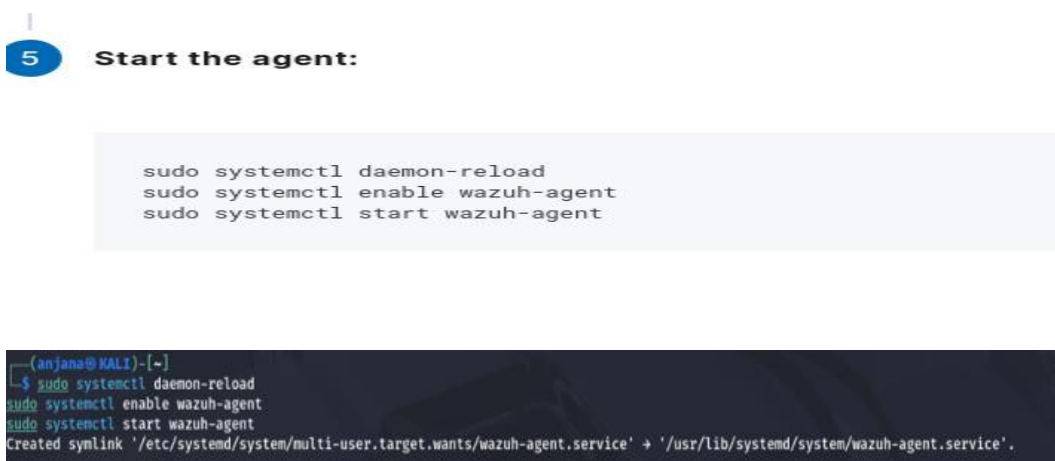
Select one or more existing groups:

default

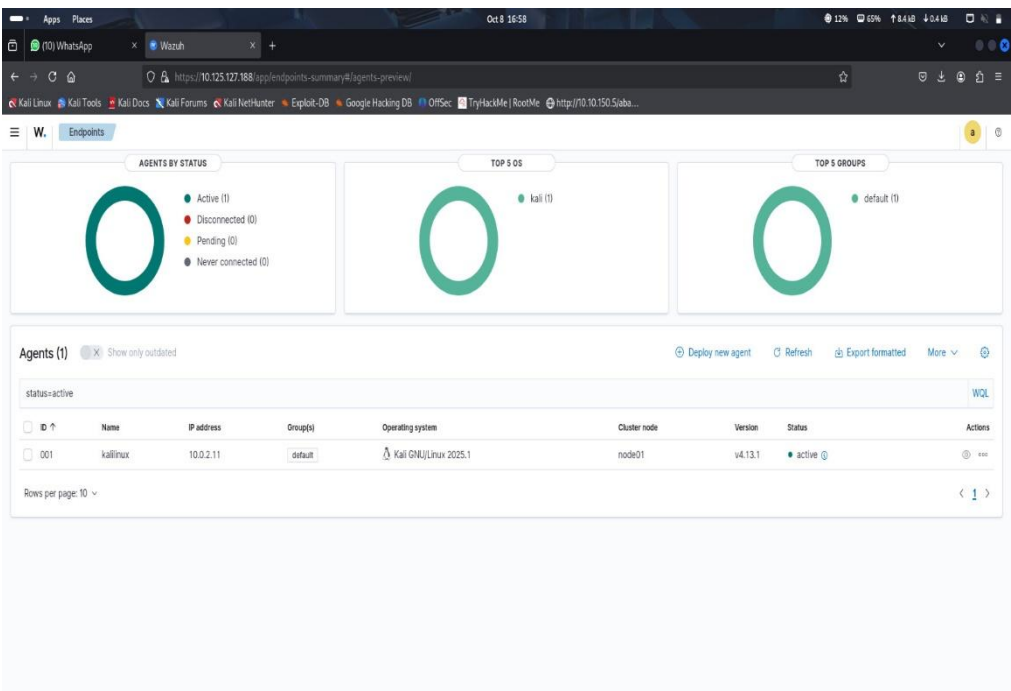
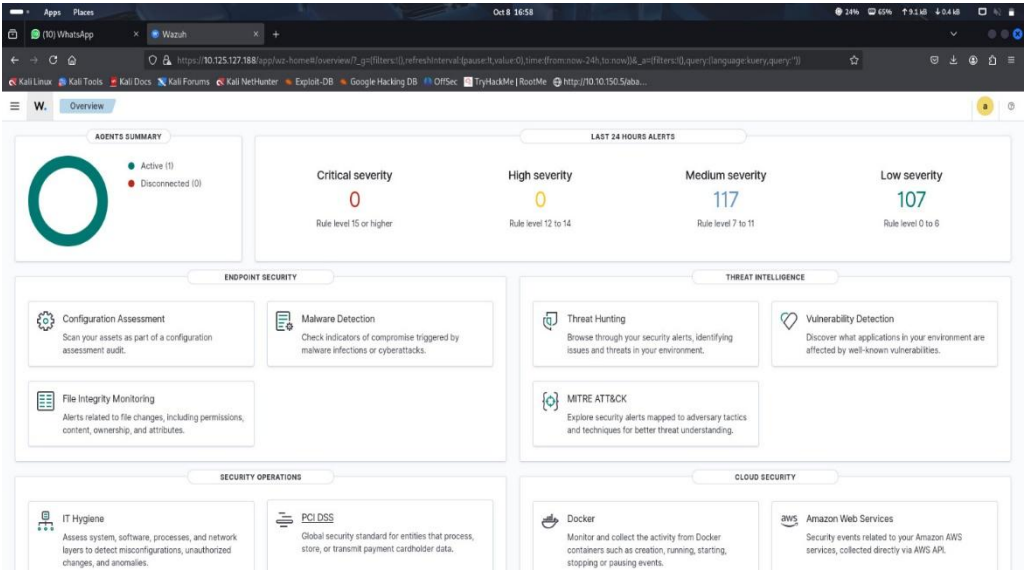
Step 4: Copy the command to run and download the agent and paste in the kali terminal.



Step 5 : Start the agent using the command

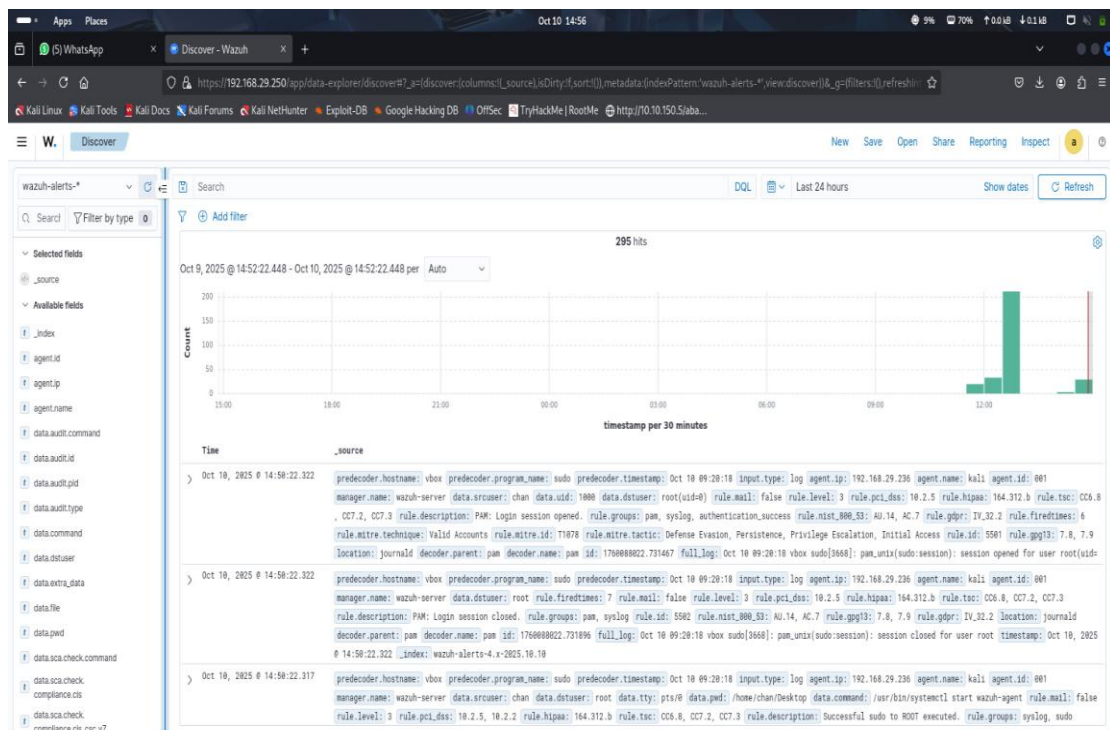


Step 6 : Created the agent and The agent is deployed successfully.The agent is active now.



4.LOG ANALYSIS

This log analysis in Wazuh shows security event monitoring from the connected agent (Kali Linux).It records activities such as user logins, command executions, and authentication results in real time.The graph displays the frequency of detected events, helping identify suspicious or abnormal system behavior.



CHALLENGES

1. Wazuh Server Installation Challenges

- ❖ Faced issues with network configuration in VirtualBox, where incorrect IP settings or NAT configurations caused connectivity failures.
- ❖ Encountered API connection timeouts and SSL certificate setup problems during the initial Wazuh configuration.
- ❖ Required additional troubleshooting to ensure the Wazuh dashboard loaded properly and connected to the backend services.

2. Wazuh Agent Installation Challenges

- ❖ Faced agent registration and connection errors, often due to mismatched manager IP addresses or firewall restrictions.
- ❖ The agent status sometimes showed as disconnected, requiring service restarts and configuration checks.
- ❖ Encountered permission-related issues while verifying log transfer and synchronization between the agent and the Wazuh server.

3. Overall Configuration Challenges

- ❖ The main difficulties involved network setup, communication between server and agent, and configuration errors.
- ❖ These were resolved by rechecking IP settings, enabling necessary services, and correctly executing agent registration commands.

CONCLUSION

The Wazuh project successfully demonstrated the implementation of an open-source Security Information and Event Management (SIEM) solution for real-time monitoring and threat detection. By integrating the Wazuh server with a Kali Linux agent, the project showcased how system activities, user behaviors, and security events can be effectively tracked and analyzed through a centralized dashboard.

Through this setup, valuable insights were gained into log analysis, intrusion detection, and endpoint monitoring, enhancing understanding of cybersecurity management and response. Despite facing configuration and connectivity challenges during installation, the project achieved its goal of establishing a functional and secure monitoring environment.

Overall, the Wazuh project proved to be an effective platform for learning and applying security monitoring, log management, and incident response techniques, reinforcing the importance of proactive cybersecurity measures in modern IT systems.

REFERENCES

Wazuh imstallation and agent setup ;

<https://youtu.be/RE27kuIe1b8?si=VfGW09MQTJNp0m9Z>